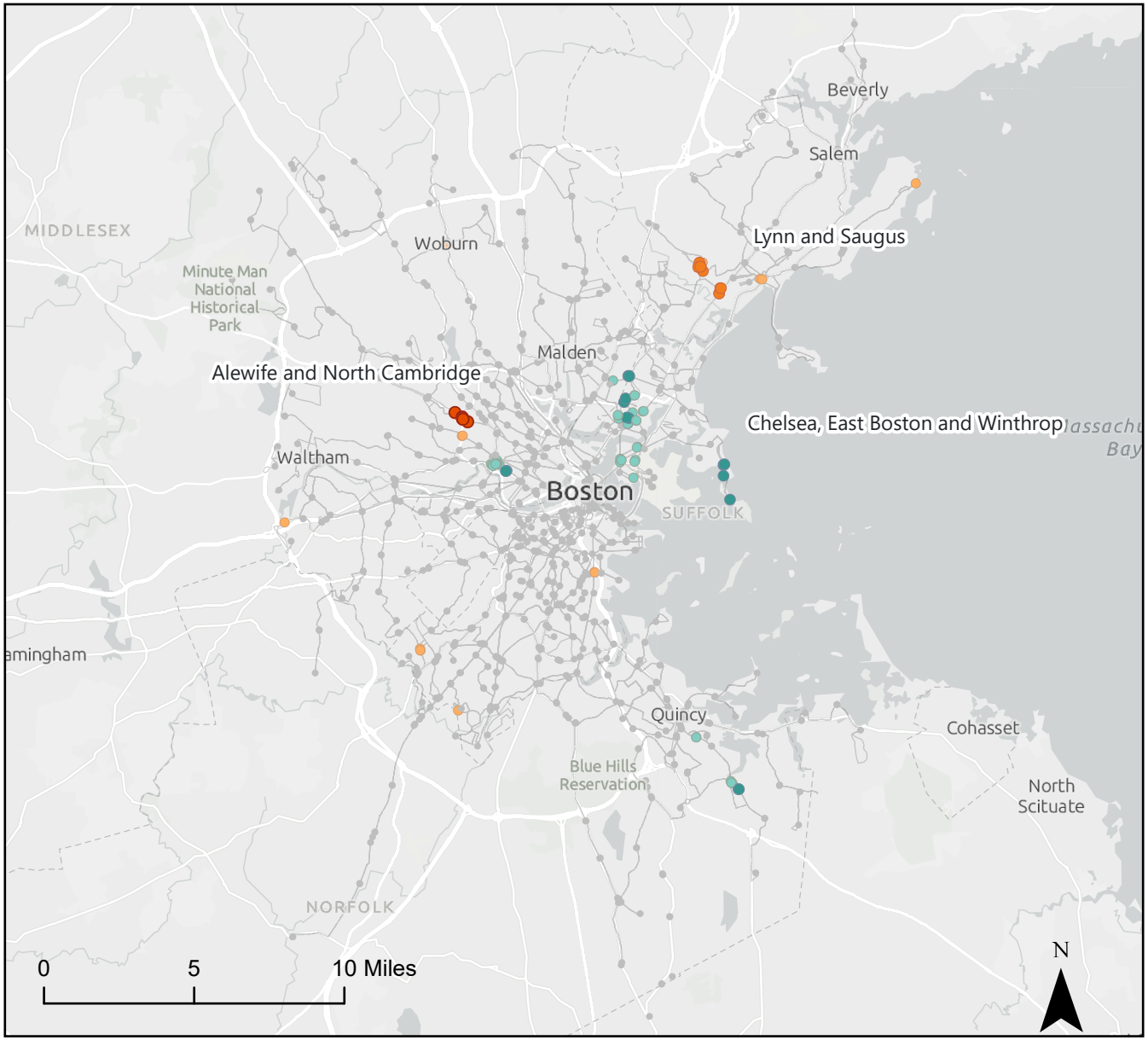
Unreliable buses cluster near Alewife and on the North Shore

Statistically significant clusters of 2025 bus on-time rates (Getis-Ord Gi*)



Clusters of on-time rate (Getis-Ord Gi*)

- Unreliable cluster, 99% confidence
- Unreliable cluster, 95% confidence
- Unreliable cluster, 90% confidence
- Not significant
- Reliable cluster, 90% confidence
- Reliable cluster, 95% confidence
- Reliable cluster, 99% confidence

How to read this map: each dot is a timepoint stop. Colored stops sit in a statistically significant cluster, where nearby stops are unusually unreliable (orange) or reliable (teal) together. Larger dots mean higher confidence. Gray stops are not part of a significant cluster.

Data: MBTA Bus Arrival Departure Times 2025; MBTA GTFS, October 2025; U.S. Census Bureau ACS 2020–2024 5-year estimates. Clusters from Hot Spot Analysis (Getis-Ord Gi*) in ArcGIS Pro on the on-time rate of 1,117 timepoint stops: 8 nearest neighbors, false discovery rate correction. Replicated in Python (PySAL) with identical results. Map: Alex Weinstein, 2026. Basemap Sources: Esri, TomTom, Garmin, (c) OpenStreetMap contributors, and the GIS User Community, Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community